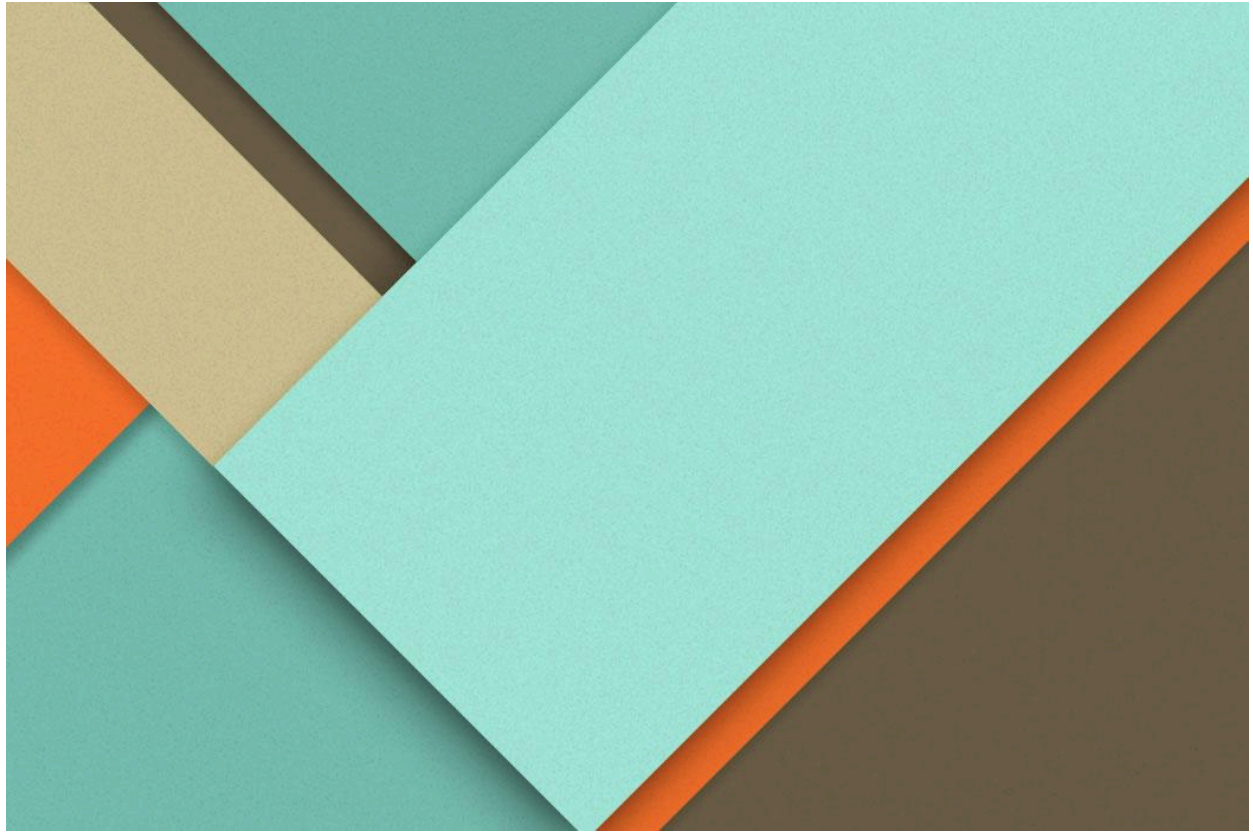




DV : StreamScope

**Netflix Content Strategy Analyzer:
Insights into Global Streaming
Trends**

Milestone 1 – Data Preparation and Cleaning

(Week 1 & 2: Requirements & Dataset Preparation)

Scope:

The goal of this milestone is to prepare and clean the Netflix Titles dataset for subsequent analysis and modeling. The dataset contains information about Netflix movies and TV shows, including metadata such as title, cast, director, country, rating, and duration.

1. Importing and Describing the Dataset

Functions Used:

- `pd.read_csv()` – Import the dataset.
- `df.shape` – Get dataset dimensions.
- `df.size` – Find total number of cells.
- `df.head()` – Preview the data.

Output:

- Shape: (8,807, 12) → 8,807 rows × 12 columns
- Total Cells: 105,684
- Columns: *show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description*

Insight:

The dataset is moderately sized and rich in categorical information, making it suitable for both descriptive analysis and machine learning.

2. Handling Duplicates

Functions Used:

- `df.shape[0]` (before and after cleaning)

- `df.drop_duplicates()` – Remove duplicate rows.

Output:

- Rows before dropping duplicates: 8,807
- Rows after dropping duplicates: 8,807
- Duplicates removed: 0

Insight:

There were no duplicate rows in the dataset.

3. Identifying Missing Values

Functions Used:

- `df.info()` – Check datatypes and non-null counts.
- `df.isnull().sum()` – Count missing values per column.

Missing Values Found In:

Column	Missing Count
Director	2634
Cast	825
Country	831
Date Added	10
Rating	4
Duration	3

Insight:

Columns such as *Director*, *Cast*, *Country*, *Date Added*, *Rating*, and *Duration* contained missing values, which required handling before analysis.

4. Handling Missing Values

Functions Used:

- `df.fillna()` – Fill missing values with placeholders.

Imputations Made:

Column	Replacement
Director	"Unknown"
Cast	"Not Available"
Country	"Unknown"
Date Added	"Unknown"
Rating	"Unrated"
Duration	"Unknown"

Insight:

After filling missing values, the dataset contained zero null entries, ensuring completeness for further processing.

5. Cleaning Text Columns

Function Used:

- `str.strip()` – Remove leading and trailing spaces from string columns (*Title, Director, Cast, Country, Description, Listed_in*).

Insight:

Ensured uniform formatting of all text-based columns for consistency.

6. Checking and Standardizing Column Formats

Functions Used:

- `df.info()` – Review datatypes and non-null counts.
- `astype()`, `str.title()` – Standardize formatting.
- `pd.to_datetime()`, `pd.to_numeric()` – Convert date and numeric columns.

Process & Output:

After verifying datatypes using `df.info()`, the final column structure was:

- 11 object columns (text)
- 1 datetime column (*date_added*)
- 1 int64 column (*release_year*)
- 1 float64 column (*duration_num*)

Insight:

All columns were standardized and formatted correctly for analysis.

7. Data Normalization & Encoding

To prepare the dataset for numerical analysis and modeling, several encoding and normalization techniques were applied.

a. Column Dropping

Function Used: `drop()`

- Dropped irrelevant columns: *rating 66min*, *rating 74min*, *rating 84min*
Insight: Removed incorrect and redundant data entries.

b. Frequency Encoding

Function Used: `value_counts()` with `map()`

- Columns: *country*, *listed_in* (Genres)
Insight: Countries and genres were replaced with numerical frequencies, allowing the model to interpret their relative importance.

c. Ordinal Encoding

Function Used: `map()` with custom order

- *release_year*: Converted to maintain chronological order.
- *rating*: Encoded by maturity level (e.g., G < PG < TV-MA).
Insight: Helped models interpret temporal and rating hierarchies.

d. One-Hot Encoding

Function Used: `get_dummies()`

- Column: *type* (Movie / TV Show)
Insight: Created two binary columns for Movies and TV Shows, ensuring proper category handling without imposing order.

e. Duration Numeric Extraction & Normalization



Functions Used: `str.extract()` and Min–Max normalization

Insight: Numeric durations (e.g., “90 min” → 90) were extracted and scaled between 0 and 1, improving comparability and model performance.

✓ Key Outcomes

- Dataset fully cleaned and standardized.
- No missing or duplicate entries remain.
- All categorical variables encoded for numerical analysis.
- Dataset ready for feature engineering and exploratory data analysis in Milestone 2.

Milestone 2 – Exploratory Data Analysis & Feature Engineering

(Week 3 & 4: Data Exploration and Feature Creation)

Project Scope & Objectives

The goal of this milestone was to explore the Netflix dataset through visual analysis and to perform feature engineering to extract deeper insights.

This stage aimed to identify content trends over time, analyze genres and ratings, and generate new variables that could support modeling and prediction tasks.

1. Netflix Content Growth Over Time-

Counted the number of titles added each year using:

```
content_per_year = df['year_added'].value_counts().sort_index()
```

Explored trends through various visualizations: Line Plot, Bar Plot, Pie Chart (for recent years), Histogram, and Scatter Plot.

Functions Used:

```
plt.plot(), plt.bar(), plt.pie(), plt.hist(), plt.scatter()
```

Insights:

- Netflix content grew rapidly after **2015**, indicating major expansion.
- Peak additions occurred between **2018–2020**.
- Growth slowed slightly post-2021, possibly due to content saturation or pandemic impact.
- Most titles are from recent years, highlighting Netflix's focus on fresh releases.

2. Genre Distribution-

Split multiple genres per title and counted their occurrences to identify the most frequent genres.

Functions Used:

```
.str.split(), .stack(), .str.strip(), .value_counts(), .plot(kind='barh')
```

Insights:

- International Movies and Dramas are the most common genres, reflecting Netflix's global appeal.
- Comedies and Documentaries are also highly popular.
- Netflix maintains a wide genre mix catering to diverse audiences.
- Horror and Reality TV have relatively less focus, indicating a stronger preference for story-based content.

3. Content Type Distribution (Movies vs. TV Shows)-

Counted titles by content type and visualized using bar and pie charts.

Functions Used:

```
.value_counts(), .plot(kind='bar'), .plot(kind='pie')
```

Insights:

- Movies make up approximately 70% of Netflix's catalog, while TV Shows account for the remaining 30%.
- This shows Netflix's roots as a movie-driven platform, gradually expanding into series for viewer engagement.

4. Ratings Distribution-

Handled missing ratings, calculated frequency counts, and visualized rating distributions and relationships with content type.

Functions Used:

```
.fillna(), .value_counts(), pd.crosstab(), .plot(stacked=True)
```

Insights:

- Most common ratings: TV-MA, TV-14, and TV-PG.
- The majority of Netflix content is rated TV-MA, indicating a focus on mature audiences.
- Movies are mostly rated R/PG-13, while TV Shows tend toward TV-14/TV-MA.

●

5. Country-Level Analysis-

Extracted and analyzed countries contributing to Netflix content.

Handled multiple country values per title and visualized the top contributors using a horizontal bar chart.

Functions Used:

`.dropna(), .str.split(), .stack(), .value_counts(), .plot(kind='barh')`

Insights:

- Top contributing countries: 🇺🇸 United States, 🇮🇳 India, 🇬🇧 United Kingdom, 🇨🇦 Canada, 🇫🇷 France.
- Reflects Netflix's strong U.S. base, with increasing content from emerging regions like India and Europe.
- Indicates a growing global content strategy.

Feature Engineering

The goal of feature engineering was to create new, informative columns that make analysis and modeling more insightful.

This process focused on transforming text-based and categorical data into structured, numeric, and analyzable forms.

1. Creating Content Length Category-

Created a custom function `get_length_category()` to classify titles based on duration.

- Movies:
 - ≤ 60 min $\rightarrow$ *Short*
 - 61–120 min $\rightarrow$ *Medium*
 - 120 min $\rightarrow$ *Long*
- TV Shows:
 - 1 Season $\rightarrow$ *Limited*
 - 2–4 Seasons $\rightarrow$ *Moderate*
 - 4 Seasons $\rightarrow$ *Long-running*

Functions Used:

`apply(), fillna()`

Output:

Added a new column, `Content_Length_Category`, classifying each title by duration.

Insights:

Most movies are Medium-length, while TV shows are primarily Limited or Moderate, reflecting Netflix's focus on shorter or limited series formats.

2. Identifying Netflix Originals-

Created a new column `is_original` to identify Netflix Originals based on whether the word “Netflix” appears in the title.

Functions Used:

`apply()` with lambda expression

Output:

Added a binary column `is_original` with values *Original* or *Licensed*.

Insights:

All titles in the dataset were categorized as Licensed, indicating Netflix’s reliance on acquired content alongside its in-house productions.

3. Yearly Growth of Netflix Originals-

Filtered “Original” titles and counted releases per year using `value_counts().sort_index()`, then visualized with a line chart.

Functions Used:

`value_counts()`, `sort_index()`, `plt.plot()`

Insights:

Only a few Netflix Originals appeared, released sporadically across 2016, 2017, 2020, and 2021, indicating limited original content in the dataset period.

4. Mapping Countries to Regions-

Mapped each country to a broader region (Asia, Europe, North America, etc.) using a predefined dictionary `region_map`.

Functions Used:

`fillna()`, `apply()`

Output:

Added a new column Region, representing the continent of origin.

Insights:

Most content originates from North America (primarily the U.S.), followed by Asia (notably India and South Korea), reflecting Netflix’s regional expansion.

5. Extracting Year from “Date Added”-

Converted the `date_added` column to datetime format and extracted the year as a new column, `Year`.

Functions Used:

`pd.to_datetime(), .dt.year`

Insights:

This enabled analysis of trends in when titles were added, revealing that most content additions occurred after 2016, aligning with Netflix’s global growth.

6. Creating Content Age Group-

Mapped each rating into simplified audience categories — *Kids*, *Teens*, *Adults*, or *Unknown* — for easier segmentation.

Functions Used:

`fillna(), map()`

Output:

Added a new column `Content_Age_Group` to represent target audience segments.

Insights:

Most Netflix content is aimed at Teens and Adults, showing the platform’s focus on mature storytelling over children’s programming.

7. Visualizations Created

- Bar Chart: Distribution of Content Length Categories (Movies mostly *Medium*, TV Shows *Limited/Moderate*).
- Bar Chart: Original vs Licensed Titles (Majority are *Licensed*).
- Line Chart: Yearly Growth of Originals (Spikes in 2016, 2017, 2020, 2021).
- Bar Chart: Audience Category Distribution (More content for *Teens* and *Adults*).
- Horizontal Bar Chart: Titles by Region (North America leads, followed by Asia).

✓ Key Outcomes

- Conducted detailed exploration of Netflix’s content trends, genres, ratings, and countries.
- Created new analytical features for deeper insights.

- 
- Found that most titles are licensed, with medium-length movies and limited-series TV shows dominating the catalog.
 - Majority of content targets Teens and Adults, emphasizing Netflix's focus on mature global storytelling.
 - The dataset is now ready for clustering, classification, and modeling in Milestone 3.

Milestone 3 – Clustering, Classification, and Global Content Analysis

(Weeks 5 & 6: Modeling and Feature Importance Analysis)

Scope:

The goal of this milestone was to perform advanced modeling and interpretation on the Netflix dataset through two major tasks:

1. Clustering and Classification: Group Netflix titles into meaningful clusters and classify content type (Movie vs. TV Show) using machine learning.
2. Global Content Analysis: Examine the key factors influencing Netflix's content availability across countries and genres using feature importance methods.

Key Concepts:

- **Clustering:** Grouping similar items without predefined categories. Netflix titles were grouped by genre, duration, and rating to uncover natural content clusters.
- **Classification:** Predicting whether a title is a *Movie* or *TV Show* based on selected features.
- **K-Means Algorithm:** Divides data into k clusters based on similarity, recalculating centroids iteratively.
- **Random Forest Model:** A supervised learning algorithm that builds multiple decision trees for accurate classification and identifies feature importance.
- **Feature Importance:** Quantifies how much each feature contributes to the model's predictive power (e.g., *duration_num* is the top indicator for differentiating Movies vs. TV Shows).

1. Clustering of Netflix Titles

Steps:

- Converted multi-genre information using **MultiLabelBinarizer**.

- Converted duration to numeric form:
 - Movies → duration in minutes.
 - TV Shows → number of seasons (approximated as 1 season = 60 min).
- Encoded *ratings* numerically.
- Combined relevant features and applied K-Means clustering to group titles into 5 clusters.

Insights from Clustering:

- Cluster sizes varied, reflecting natural content segmentation.
Representative clusters included:
 - Cluster 0: Multi-genre TV Shows (Crime, International, Spanish-language).
 - Cluster 1: Long movies, often dramas or international films.
 - Cluster 2: Shorter movies, mainly dramas and international titles.
 - Cluster 3: Stand-up comedy specials.
 - Cluster 4: Limited TV series (1–2 seasons), often crime or drama-based.

Visualizations:

- Bar Chart: Number of titles per cluster.
- Pie Chart: Percentage distribution of titles across clusters.

Insight:

The clustering effectively revealed distinct content groups that align with genre and duration patterns, showing Netflix's balanced mix of short films, long dramas, and limited TV series.

2. Classification of Content Type (Movie vs TV Show)

Steps:

- Selected four key features for classification:
 - *listed_in* (Primary Genre)
 - *rating* (Content Rating)
 - *Content_Age_Group* (Audience Segment)
 - *duration_num* (Duration in Minutes or Seasons)

- Encoded categorical variables into numerical form.
- Trained a Random Forest Classifier and evaluated using accuracy, confusion matrix, and feature importance.

Model Performance:

- Accuracy: 99.94%
- Confusion Matrix:
 - True Positives (TV Shows): 547
 - True Negatives (Movies): 1214
 - False Positives: 0
 - False Negatives: 1

Insight:

The model demonstrated near-perfect classification, confirming that these selected features clearly differentiate movies and TV shows.

3. Feature Importance (Classification Model)

Feature	Importance Score
duration_num	0.778
listed_in	0.198
rating	0.024
Content_Age_Group	0.001

Insights:

- Duration (duration_num) was the most influential feature, serving as the primary differentiator between Movies and TV Shows.
- Genre (listed_in) was the second most important factor, refining the model's accuracy.
- Rating and Content_Age_Group contributed minimally.

Visualization:

A horizontal bar chart illustrated that *duration_num* dominated feature importance, followed by genre.

4. Global Content Availability Analysis

Steps:

- Imported and cleaned the extended dataset (*netflix_titles_featured.csv*).
- Filled missing values in *country*, *listed_in*, and *date_added*.
- Converted *date_added* to datetime format and handled multiple countries/genres per title using *explode()*.
- Grouped data by *country* and *genre* to count total titles per combination.
- Created analytical columns:
 - *is_top_country* → 1 if the country is in the top 10 by title count.
 - *is_top_genre* → 1 if the genre is among the top 10 by frequency.
 - *duration* → numeric extraction from *duration*.
 - *rating* → numeric codes.
 - *content_age_years* → (2025 – *release_year*).
- Selected features: *duration*, *rating*, *content_age_years*, *is_top_genre*
- Target variable: *is_top_country*.
- Trained a Logistic Regression model to identify factors affecting top-country content availability.

Functions Used:

pd.read_csv(), *fillna()*, *assign()*, *explode()*, *groupby()*, *astype()*, *str.extract()*, *LogisticRegression()*

Output:

Top Country–Genre Combinations:

- India – International Movies (864)
- United States – Dramas (835)
- United States – Comedies (680)

- India – Dramas (662)
- United States – Documentaries (511)

Insights:

The U.S. and India lead Netflix’s catalog, primarily in dramas and comedies.

Documentaries and international movies show strong cross-regional appeal, emphasizing Netflix’s global, multicultural content strategy.

5. Feature Importance Analysis (Global Model)

Steps:

- Fitted the Logistic Regression model and extracted coefficients to measure feature significance.
- Visualized results using a bar chart for interpretability.

Functions Used:

`LogisticRegression().fit(), clf.coef_, pandas.DataFrame(), matplotlib.pyplot.bar()`

Output (Coefficients):

Feature	Coefficient
is_top_genre	-0.6077
rating	-0.0761
content_age_years	+0.0065
duration	+0.0003

Insights:

- Genre popularity strongly influences Netflix’s country-level catalog.
- Newer content is more widely distributed across countries.
- Rating and duration play minor roles in determining regional availability.

- Top countries showcase greater genre diversity, highlighting Netflix's focus on localized yet varied content.

6. Content Type Distribution by Country

Steps:

- Exploded the *country* column to count the number of Movies and TV Shows per country.
- Calculated percentages and visualized the results using a stacked bar chart.

Functions Used:

`groupby()`, `size()`, `merge()`, `assign()`, `pivot()`, `matplotlib.pyplot.bar()`

Output (Sample):

Country	Movie %	TV Show %
Afghanistan	100.0	0.0
Argentina	78.0	22.0
Australia	58.8	41.2
Austria	91.7	8.3

Insights:

Most countries have a higher proportion of Movies than TV Shows, illustrating Netflix's strong movie-focused distribution globally, while TV content is steadily expanding.

Visualization:

A stacked bar chart (Movies – Blue, TV Shows – Orange) represented Netflix's content composition by country.

Key Outcomes

- Successfully performed both unsupervised (clustering) and supervised (classification, regression) analyses.

- Clustering: Revealed logical content groupings by genre, duration, and rating.
- Classification: Achieved 99.94% accuracy in predicting Movies vs. TV Shows using Random Forest.
- Feature Importance: Identified *duration_num* as the strongest differentiator for content type.
- Global Analysis: Found *genre* and *content recency* as key drivers of international content distribution.
- Country Insights: The U.S. and India dominate Netflix's library, with genre diversity reflecting global expansion.

Milestone 4 – Power BI Dashboard & Visualization Analysis

Scope

The goal of this milestone was to design and develop an interactive Power BI dashboard using the cleaned Netflix dataset. The dashboard provides visual insights into Netflix’s content library — including growth trends, genre distribution, content ratings, and geographical patterns.

This analysis helps identify which types of content dominate Netflix, how its catalog has evolved over time, and where the majority of titles are produced.

1: Loading the Dataset

- Imported the cleaned netflix_titles.csv dataset into Power BI.
- Verified all column names, data types, and value consistency.

Key Columns Used:

type, title, country, release_year, date_added, listed_in, rating, duration, primary_genre, year_added

Insight:

The dataset structure was well-optimized for visualization — containing relevant fields for filtering, aggregation, and time-based analysis.

2: Data Transformation (Power Query)

All necessary data cleaning was performed using Power Query Editor before visualization.

Transformations Applied:

- Replaced null values with placeholders (e.g., “Unknown”, “Not Available”).

- Converted data types for accuracy:
 - `date_added` → Date
 - `release_year` → Whole Number
- Extracted new fields:
 - **Year Added** (from `date_added`)
 - **Primary Genre** (from `listed_in`)
- Trimmed extra spaces and cleaned text formatting using *Transform* → *Format* → *Trim/Clean*.
- Verified there were no empty or invalid values remaining.
- Removed columns that were not required in the analysis like directors, cast etc
- Applied all transformations and loaded the finalized dataset into Power BI.

Insight:

The dataset was made analysis-ready with standardized formats and derived fields for time and genre-based visualization.

3: DAX Measures Creation

Custom DAX measures were created to summarize and analyze key performance indicators (KPIs) in the dashboard.

Measure	Formula	Purpose
Total Titles	<code>COUNTROWS(netflix_titles)</code>	Total number of titles available on Netflix.
Total Movies	<code>CALCULATE(COUNTROWS(netflix_titles), netflix_titles[type]="Movie")</code>	Count of all movies.
Total TV Shows	<code>CALCULATE(COUNTROWS(netflix_titles), netflix_titles[type]="TV Show")</code>	Count of all TV shows.

Average Titles per Year	<code>AVERAGEX(VALUES(netflix_titles[release_year]), CALCULATE(COUNTROWS(netflix_titles)))</code>	Average number of titles released per year.
Movies vs TV Ratio	<code>DIVIDE([Total Movies], [Total TV Shows], 0)</code>	Ratio comparison between movies and TV shows.

Insight:

These DAX measures served as the foundation for dynamic metrics and visual storytelling within the dashboard.

4: Adding KPI Cards

To provide at-a-glance performance metrics summarizing Netflix's key statistics.

KPI Cards Created:

- Total Titles: 9K
- Total Movies: 6K
- Most Common Genre: *Dramas*
- Average Titles per Year: 118.91
- Movies vs TV Show Ratio: 2.29

Insight:

The KPI section instantly communicates Netflix's dominant content types and overall library size.

5: Adding Slicers (Interactive Filters)

Slicers were added to enhance dashboard interactivity and enable dynamic exploration.

Slicers Added:

1. Year Added – Filter by year titles were added to Netflix.
2. Primary Genre – Filter content by specific genres.
3. Show ID – Select specific titles using their unique IDs.

4. Type – Toggle between Movies and TV Shows.
5. Rating – Filter by maturity rating (e.g., TV-MA, PG-13).
6. Title – View detailed information for selected titles.
7. Country – Filter titles based on country of origin.
8. Duration – Explore based on runtime or season count.

Insight:

These filters enabled user-driven data exploration, allowing users to analyze Netflix's catalog from multiple perspectives — year, country, genre, and content type.

6: Visualizations and Insights

1] Growth of Netflix Content Over the Years

Visual Type: Line Chart

Fields Used:

- X-axis → `release_year`
- Y-axis → Count of `show_id`

Insight:

There is a consistent rise in Netflix titles post-2010, with a significant spike between 2015–2020, highlighting Netflix's global expansion and large-scale content acquisition phase.

2] Top 10 Genres on Netflix

Visual Type: Horizontal Bar Chart

Fields Used:

- Y-axis → `primary_genre`
- X-axis → Count of `show_id`
Filter: Top 10 by title count

Insight:

Dramas and Comedies dominate Netflix's catalog, followed by Documentaries and Action genres — reflecting a balanced mix of storytelling formats across categories.

3 Country Distribution

Visual Type: Map

Fields Used:

- Location → **country**
- Size → Count of **show_id**

Insight:

The United States, India, and the United Kingdom produce the majority of Netflix titles, showcasing Netflix's strong investment in North American and Asian markets.

4 Netflix Content by Rating

Visual Type: Pie Chart

Fields Used:

- Legend → **rating**
- Values → Count of **show_id**

Insight:

Most Netflix content is rated TV-MA and TV-14, indicating a primary focus on mature and teen audiences.

5 Movie vs TV Show Trend by Year

Visual Type: Line Chart

Fields Used:

- X-axis → **release_year**
- Y-axis → Count of **show_id**

- Legend → type

Insight:

Both Movies and TV Shows have increased steadily since 2010, with Movies maintaining a slight lead in total volume over time.

♦ Interactive Slicers

Three interactive slicers — Country, Release Year (Dropdown), and Type (Movie/TV Show) — were added and linked to all dashboard visuals.

These slicers allow users to filter and explore Netflix content dynamically by region, release timeline, and content type, ensuring a more customized and interactive data analysis experience across the entire dashboard.

✓ Key Insights

- Netflix's content catalog has expanded massively since 2010, reflecting its global growth strategy.
- Movies dominate the platform (~70%), though TV Shows continue to grow steadily.
- Top Genres: Drama, Comedy, and Documentary account for the majority of Netflix content.
- Audience Targeting: Most titles are rated TV-MA or TV-14, indicating a strong focus on mature audiences.
- Geographical Insights: The USA, India, and UK remain top contributors to Netflix's library.
- Dashboard Interactivity: Global slicers for Country, Release Year, and Type enhance data exploration across all visuals, making the dashboard highly dynamic and user-friendly.



